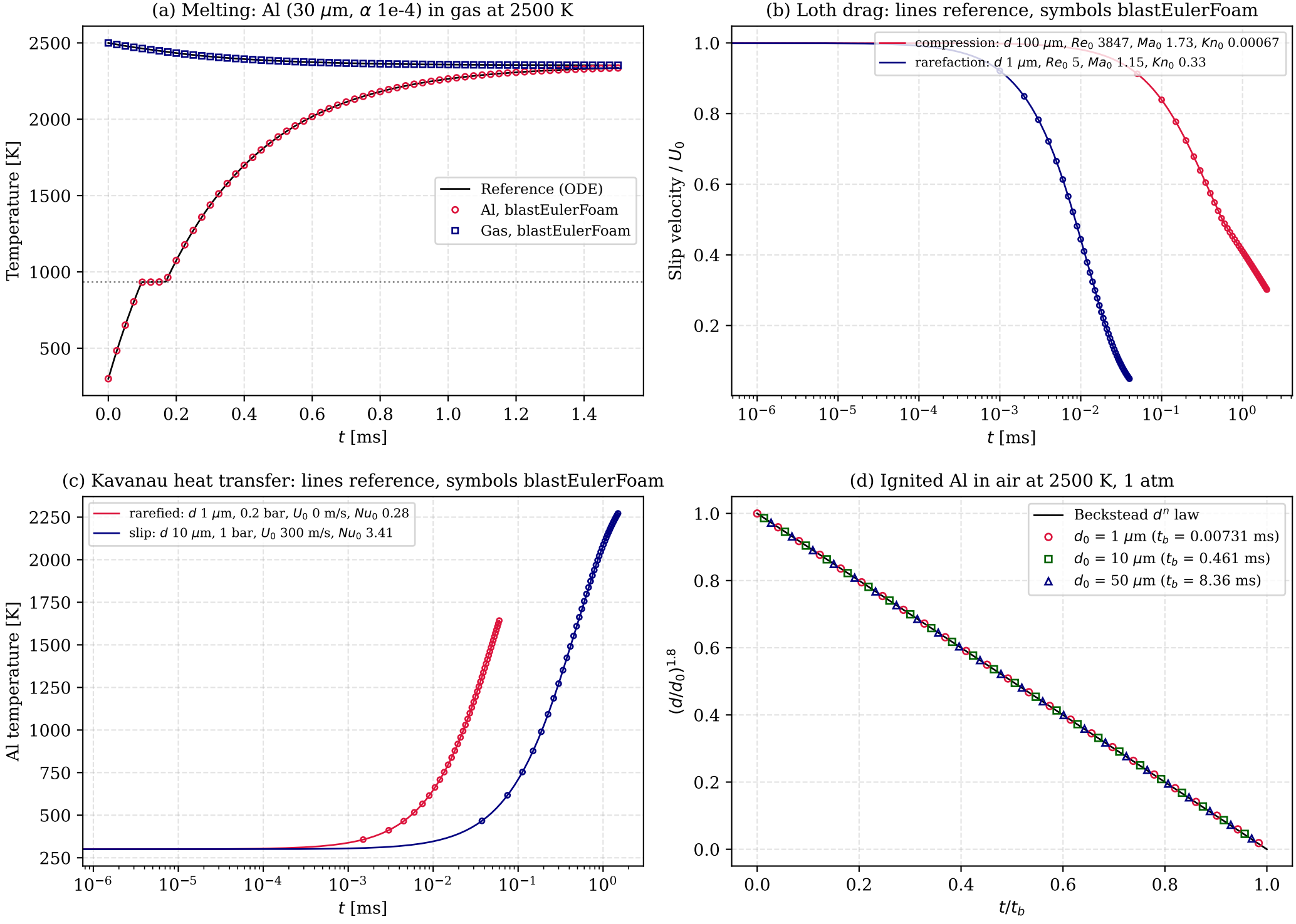
(a) Melting: Al (30 μm, α 1e-4) in gas at 2500 K
Temperature [K]
t [ms]
Reference (ODE)
Al, blastEulerFoam
Gas, blastEulerFoam
(b) Loth drag: lines reference, symbols blastEulerFoam
Slip velocity / U0
t [ms]
compression: d 100 μm, Re0 3847, Ma0 1.73, Kn0 0.00067
rarefaction: d 1 μm, Re0 5, Ma0 1.15, Kn0 0.33
(c) Kavanau heat transfer: lines reference, symbols blastEulerFoam
Al temperature [K]
t [ms]
rarefied: d 1 μm, 0.2 bar, U0 0 m/s, Nu0 0.28
slip: d 10 μm, 1 bar, U0 300 m/s, Nu0 3.41
(d) Ignited Al in air at 2500 K, 1 atm
(d/d0)^1.8
t/tb
Beckstead d^n law
d0 = 1 μm (tb = 0.00731 ms)
d0 = 10 μm (tb = 0.461 ms)
d0 = 50 μm (tb = 8.36 ms)